

What is your capstone agent designed to do, and what safety or reliability risks could arise in its intended workflow?

- My agent is designed to plan an itinerary for users. It is not designed to book flights, tickets, or reservations. While this is an incredibly powerful feature, there are too many risks associated with it for the current scope.
- Some more applicable risks are that the agent may capture the wrong data and provide it to the user. If the user fails to verify it, they may be missing crucial information, such as difficulties of hikes, opening hours, etc.

What guardrails will you use to constrain the behavior of your agent? Consider guardrails such as input checks, output constraints, source verification, tool access limits, escalation rules, or runtime monitoring.

- Every output from the agent will be verified by another agent. This ensures double verification. Furthermore, the agent's tools will be restricted to just search so that it isn't able to make any purchases on behalf of the user.

What evaluation metrics will you use to assess the performance of your agent? You may include metrics related to correctness, groundedness, calibration, safety, latency, fallback success, escalation rate, or other measures that fit your use case.

- I will use the 7 metrics as described in the prelecture readings: accuracy, robustness, calibration, bias, fairness, toxicity, and efficiency. It will be tested in several scenarios that test each one of these.

Under what conditions should your system require human intervention rather than allowing the agent to proceed autonomously? Consider factors such as ambiguity, confidence, risk level, action impact, or policy sensitivity.

- There will be three modes the agent can run in. The first is fully autonomous where it plans any trip with no restrictions in details. This is useful when users don't care where they go or are looking for something completely new to their usual vacations. The second is similar but with some restrictions such as what season or what continent to visit. The last mode is the highest restrictions with the most amount of details such as flying in on X and leaving on Y. The agent will only have access to the web to search for critical information. The agent can not take on any actions such as booking flights. It can only provide details and form an itinerary.

How will guardrails, evaluation, and human intervention work together as part of an overall safety strategy for your capstone agent?

- With the guardrails, evaluation, and human intervention, the agent will be confined to minimal harmful action capabilities, have its responses thoroughly reviewed and fact checked, and always warns the user to verify before purchasing or reserving anything.

What trade-offs did you consider in your design, particularly in terms of autonomy, reliability, efficiency, and oversight?

- With more restrictions, guardrails, and testing, it's going to take longer and cost more. However, the upsides are that the agent will provide more reliable responses. Given that people tend to spend a few days planning, there isn't a strong urge to have super fast response time. This is a good trade off.
- In terms of disabling the agent from booking things on behalf of the the user, there are just too many security concerns involved with this. The risks of this feature are simply too high for the benefits it provides.

How does your plan support safer and more dependable deployment of your agent in a real-world context?

- Every suggestion that the agent provides is passed through a verification agent that checks for accuracy and if it meets all the criteria listed by the user. This enables results that are both accurate and applicable. There were tradeoffs considered that allow the agent to be useful while mitigating strong risks. This allows it to have a real impact on users while being secured at the same time.